

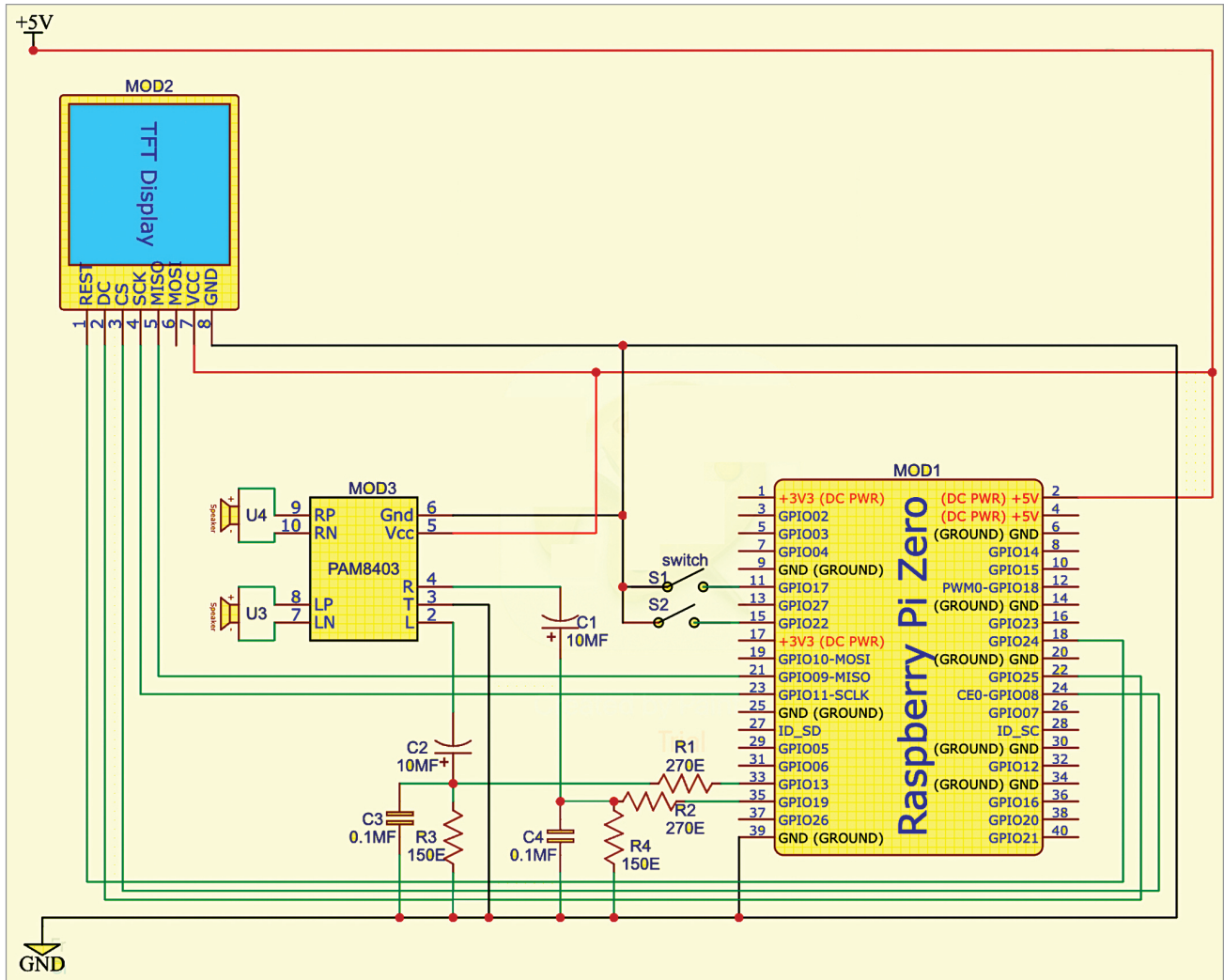


Fig. 2: Circuit diagram

sentence, press the button again. The device processes your speech and converts it into Hindi, producing clear and fluent Hindi output. Each step of the process is displayed on the screen, including:

- Recording the speech
- Converting the recorded voice to text
- Translating the text into speech using Google Translate and Google Text-to-Speech

To switch to 'Hindi to English,' press the 'hi-en' button, wait for the message 'Recording started (Hindi to English)' to appear and proceed with your input.

This process is resource-intensive for the Pi Zero board. To manage the workload, the Pi Zero is overclocked to 1.3GHz with a quality metallic heatsink on the processor. A stable internet connection



Fig. 3: USB microphone

is essential; otherwise, the process may take longer.

Final setup and testing

Fig. 4 shows the Raspberry Pi Zero pins before connection. Connect the components to the Raspberry Pi as per the circuit diagram (refer to Fig. 2). Attach the microphone to the USB port for sound input used in translation.

To start the system at boot:

- Add the program to the user's .profile file to run on boot.
- Redirect errors to /dev/null to suppress non-critical messages.
- Ensure the Pi Zero logs into the